

LAB SHEET -2

Objective:

- To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

```
1 CREATE TABLE CLASS (  
2     id NUMBER PRIMARY KEY,  
3     name VARCHAR2(50)  
4 );  
5
```

```
6 INSERT INTO CLASS VALUES (1, 'Bravo');  
7 INSERT INTO CLASS VALUES (2, 'Alex');  
8 INSERT INTO CLASS VALUES (4, 'Cheng');
```

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

```

1  CREATE TABLE EMPLOYEES (
2      Employee_Id NUMBER PRIMARY KEY,
3      First_Name VARCHAR2(50),
4      Last_Name VARCHAR2(50),
5      Email VARCHAR2(50),
6      Phone_Number VARCHAR2(20),
7      Hire_Date DATE,
8      Job_Id VARCHAR2(20),
9      Salary NUMBER,
10     Commission_Pct NUMBER,
11     Manager_Id NUMBER,
12     Department_Id NUMBER
13 );

```

DDL Questions (Data Definition Language):

- How do you create a table named CLASS with appropriate data types?
- How do you create a table named EMPLOYEES with the specified attributes?
- How do you create a table named Client_master with the specified attributes?
- What SQL statement would you use to rename the table Client_master to Client12?
- How do you display the bal_due column heading as "BALANCE"?

```

1  CREATE TABLE Client_master (
2      ClientNO VARCHAR2(10) PRIMARY KEY,
3      Name VARCHAR2(50),
4      City VARCHAR2(50),
5      bal_due NUMBER
6  );

```

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,'Rahul');
- COMMIT;
- UPDATE class SET name = 'Abhijeet' WHERE id='5';
- SAVEPOINT A;
- INSERT INTO class VALUES (6,'Chris');

- SAVEPOINT B;
- INSERT INTO class VALUES (7,'Bravo');
- SAVEPOINT C;
- SELECT * FROM class;
- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

```

11 * COMMIT;
12 * UPDATE CLASS SET name='Abhijeet' WHERE id=5;
13 * SAVEPOINT A;
14 * INSERT INTO CLASS VALUES (6,'Chris');
15 * SAVEPOINT B;
16 * INSERT INTO CLASS VALUES (7,'Bravo');
17 * SAVEPOINT C;
18 * SELECT * FROM CLASS;
19 * ROLLBACK TO B;
20 * SELECT * FROM CLASS;
21 * ROLLBACK TO A;

```

Part B: Retrieving Records (EMPLOYEES Table)

- Find out the employee id, names, and salaries of all the employees.

```

17 * SELECT Employee_Id, First_Name, Salary
18 FROM EMPLOYEES;

```

- List out the employees who work under manager 100.

```

21 * SELECT *
22 FROM EMPLOYEES
23 WHERE Manager_Id = 100;

```

- Find the names of the employees who have a salary greater than or equal to 4800.

```

27 * SELECT First_Name
28 FROM EMPLOYEES
29 WHERE Salary >= 4800;

```

- List out the employees whose last name is 'AUSTIN'.

```
32 • SELECT *
33     FROM EMPLOYEES
34     WHERE Last_Name = 'AUSTIN';
35
```

- Find the names of the employees who work in departments 60, 70, and 80.

```
38 • SELECT *
39     FROM EMPLOYEES
40     WHERE Department_Id IN (60,70,80);
```

- Display the unique Manager_Id.

```
43 • SELECT DISTINCT Manager_Id
44     FROM EMPLOYEES;
```

Part C: Updating Records (Client_master Table)

- Insert five records into the Client_master table.

```
10 • INSERT INTO Client_master VALUES ('C101','Aman','Delhi',6000);
11 • INSERT INTO Client_master VALUES ('C102','Riya','Noida',3000);
12 • INSERT INTO Client_master VALUES ('C103','Rahul','Mumbai',7000);
13 • INSERT INTO Client_master VALUES ('C104','Neha','Pune',2000);
14 • INSERT INTO Client_master VALUES ('C123','Karan','Delhi',4500);
```

- Find the names of clients whose bal_due > 5000.

```
18 • SELECT Name
19     FROM Client_master
20     WHERE bal_due > 5000;
```

- Change the bal_due of ClientNO "C123" to Rs. 5100.

```
24 • UPDATE Client_master
25     SET bal_due = 5100
26     WHERE ClientNO = 'C123';
```